

SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0716

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%
Software: Cisco Packet Tracer, Wireshark

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

1. Problem Identification

A workstation is configured with the IPv4 address 192.168.10.65, subnet mask 255.255.255.192 (/26), and default gateway 192.168.10.1. However, the workstation cannot communicate with other devices in the subnet 192.168.10.0/26. The objective is to identify the configuration error, verify the addressing details, and optimize the network configuration for successful communication.

2. Given Configuration

Parameter	Configured Value
IP Address	192.168.10.65
Subnet Mask	255.255.255.192 (/26)
Default Gateway	192.168.10.1
Expected Subnet	192.168.10.0/26

3. Error Analysis

Host Bits

$$32 - 26 = 6$$

Block Size

$$256 - 192 = 64$$

Subnet ranges are:

Subnet 1

192.168.10.0 – 192.168.10.63

Subnet 2

192.168.10.64 – 192.168.10.127

The configured IP address

192.168.10.65

belongs to

192.168.10.64/26

The configured default gateway

192.168.10.1

belongs to

192.168.10.0/26

Therefore,

the workstation and gateway belong to different subnets.

This is the primary configuration error.

4. Debugging Process

Step 1

Verify the subnet mask.

255.255.255.192 (/26) is correct.

Step 2

Identify the subnet of the configured IP.

192.168.10.65 belongs to

192.168.10.64/26

Step 3

Verify the gateway.

192.168.10.1 belongs to

192.168.10.0/26

Step 4

Compare both subnets.

Workstation

192.168.10.64/26

Gateway

192.168.10.0/26

Mismatch detected.

Communication fails because the host and gateway are in different networks.

5. Step-by-Step Calculation

Current Subnet

192.168.10.64/26

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Valid Host Range

192.168.10.65 – 192.168.10.126

Expected Subnet

192.168.10.0/26

Network Address

192.168.10.0

Broadcast Address

192.168.10.63

Valid Host Range

192.168.10.1 – 192.168.10.62

Host Bits

6

Total Addresses

$2^6 = 64$

Usable Hosts

$64 - 2 = 62$

6. Optimization

To communicate with the subnet

192.168.10.0/26

the workstation IP address must belong to the same subnet as the gateway.

Recommended Workstation IP

192.168.10.2

Default Gateway

192.168.10.1

Subnet Mask

255.255.255.192

Both devices now belong to

192.168.10.0/26

7. Optimized Configuration

Parameter	Optimized Value
IP Address	192.168.10.2
Subnet Mask	255.255.255.192
Default Gateway	192.168.10.1
Network Address	192.168.10.0
Broadcast Address	192.168.10.63
Valid Host Range	192.168.10.1 – 192.168.10.62
Usable Host Addresses	62

8. Final Result

Parameter	Final Output Value
Original IP Address	192.168.10.65
Original Gateway	192.168.10.1
Detected Error	Host and Gateway belong to different subnets
Corrected IP Address	192.168.10.2
Subnet Mask	255.255.255.192 (/26)
Correct Gateway	192.168.10.1
Network Address	192.168.10.0
Broadcast Address	192.168.10.63
Valid Host Range	192.168.10.1 – 192.168.10.62
Usable Host Addresses	62

Detected Error

Host and Gateway belong to different subnets

9. Analysis & Justification

The communication failure occurred because the workstation IP address belonged to the subnet 192.168.10.64/26, while the configured default gateway belonged to the subnet 192.168.10.0/26. Devices must be in the same subnet as their default gateway for successful local communication. By assigning the workstation a valid IP address within the 192.168.10.0/26 network while retaining the existing gateway, the configuration is optimized. The corrected network supports 62 usable host addresses and enables reliable communication with other systems in the subnet.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

1. Problem Identification

Two devices are configured with the IPv6 addresses 2001:db8::ff00:42:8329/64 and 2001:db8:0:0:1::1/64. Although both devices are connected to the same physical network, they are unable to communicate. The objective is to debug the IPv6 configuration by verifying the network prefixes, identifying the addressing mismatch, and optimizing the addressing scheme to enable successful communication.

2. Given Configuration

Device	IPv6 Address / Prefix
Device A	2001:db8::ff00:42:8329/64
Device B	2001:db8:0:0:1::1/64
Prefix Length	/64

3. Error Analysis

Expand Device A Address

2001:0db8:0000:0000:0000:ff00:0042:8329

Expand Device B Address

2001:0db8:0000:0000:0001:0000:0000:0001

Determine Network Prefix

Device A

2001:0db8:0000:0000::/64

Device B

2001:0db8:0000:0000::/64

Both devices have the same /64 network prefix.

However, Device B uses a different Interface Identifier (Host ID), and if communication still fails, the issue is likely related to incorrect IPv6 configuration such as gateway configuration, Duplicate Address Detection (DAD), Neighbor Discovery Protocol (NDP), or interface settings rather than the network prefix itself.

4. Debugging Process

Step 1

Expand both IPv6 addresses.

Completed successfully.

Step 2

Compare the first 64 bits.

Both devices have identical network prefixes.

Step 3

Verify subnet membership.

Device A

2001:db8:0000:0000::/64

Device B

2001:db8:0000:0000::/64

Both belong to the same subnet.

Step 4

Identify the possible issue.

No prefix mismatch exists.

Possible causes include:

- Incorrect interface configuration.
- Disabled IPv6.
- Neighbor Discovery failure.
- Duplicate Address Detection failure.
- Incorrect default gateway configuration.

5. Step-by-Step Calculation

IPv6 Address Length

128 bits

Prefix Length

64 bits

Host Bits

128 – 64

= 64

Total Available Host Addresses

2^{64}

=

18,446,744,073,709,551,616

6. Optimization

Assign IPv6 addresses within the same subnet using the same /64 prefix.

Recommended Configuration

Device A

2001:db8::10/64

Device B

2001:db8::20/64

Network Prefix

2001:db8::/64

Both devices now belong to the same IPv6 network with unique interface identifiers.

7. Optimized Configuration

Parameter / Field	Optimized Value
Device A IPv6 Address	2001:db8::10/64
Device B IPv6 Address	2001:db8::20/64
Network Prefix	2001:db8::/64
Host Bits	64
Available Host Addresses	18,446,744,073,709,551,616

8. Final Result

Parameter	Final Output Value
Device A Prefix	2001:db8::/64
Device B Prefix	2001:db8::/64
Same Subnet	Yes
Addressing Mismatch	No Prefix Mismatch
Possible Error	Interface or IPv6 Configuration
Optimized Device A	2001:db8::10/64
Optimized Device B	2001:db8::20/64
Available Host Addresses	18,446,744,073,709,551,616

Addressing Mismatch

No Prefix Mismatch

Possible Error

Interface or IPv6 Configuration

Available Host Addresses

18,446,744,073,709,551,616

9. Analysis & Justification

The expanded IPv6 addresses confirm that both devices belong to the same /64 network prefix. Therefore, the communication failure is not caused by an addressing or prefix mismatch. The issue is more likely related to interface configuration, Neighbor Discovery Protocol (NDP), Duplicate Address Detection (DAD), or default gateway configuration. Assigning simple, unique IPv6 addresses within the same /64 subnet and verifying the IPv6 interface configuration ensures reliable communication while maintaining the standard IPv6 addressing scheme. The corrected subnet provides 2^64 possible host addresses, making it suitable for large-scale enterprise and IoT deployments.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.

- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

1. Problem Identification

An organization has allocated the network 192.168.1.0/24 into multiple departmental subnets using the subnet masks /26, /27, and /28. Some departments experience insufficient IP addresses, whereas others have many unused addresses. The objective is to analyze the existing subnet allocation, identify inefficient address utilization, locate subnet allocation mistakes, and optimize the network using Variable Length Subnet Masking (VLSM).

2. Given Configuration

Parameter	Given Specification
Original Network	192.168.1.0/24
Department 1 Subnet	/26 (64 addresses)
Department 2 Subnet	/27 (32 addresses)
Department 3 Subnet	/28 (16 addresses)
Optimization Scheme	Variable Length Subnet Masking (VLSM)

3. Error Analysis

Existing subnet allocation

Subnet 1

Prefix

/26

Network Address

192.168.1.0

Broadcast Address

192.168.1.63

Valid Host Range

192.168.1.1 – 192.168.1.62

Usable Hosts

62

Subnet 2

Prefix

/27

Network Address

192.168.1.64

Broadcast Address

192.168.1.95

Valid Host Range

192.168.1.65 – 192.168.1.94

Usable Hosts

30

Subnet 3

Prefix

/28

Network Address

192.168.1.96

Broadcast Address

192.168.1.111

Valid Host Range

192.168.1.97 – 192.168.1.110

Usable Hosts

14

Analysis of Address Utilization

Department using /26

Available Hosts

62

Unused Hosts

High if fewer than 62 devices are connected.

Department using /27

Available Hosts

30

Unused Hosts

Moderate depending on the number of connected devices.

Department using /28

Available Hosts

14

If the department requires more than 14 devices, the subnet becomes insufficient.

4. Debugging Process**Step 1**

Verify each subnet boundary.

No subnet overlap is detected.

Step 2

Verify broadcast addresses.

All broadcast addresses are correct.

Step 3

Verify host ranges.

All host ranges are valid.

Step 4

Identify inefficient allocation.

Large departments should receive larger subnets.

Smaller departments should receive smaller subnets.

The existing allocation may waste addresses if department sizes do not match the assigned subnet sizes.

Step 5

Identify optimization requirement.

Redistribute address space according to the actual number of devices using VLSM.

5. Step-by-Step Calculation

Subnet /26

Host Bits

6

Total Addresses

64

Usable Hosts

62

Subnet /27

Host Bits

5

Total Addresses

32

Usable Hosts

30

Subnet /28

Host Bits

4

Total Addresses

16

Usable Hosts

14

Remaining Address Space

192.168.1.112 – 192.168.1.255

Available for future subnet allocation using VLSM.

6. Optimization

Recommended VLSM Allocation

Allocate the largest subnet to the department with the highest number of users.

Example

Largest Department

192.168.1.0/26

62 usable hosts

Medium Department

192.168.1.64/27

30 usable hosts

Small Department

192.168.1.96/28

14 usable hosts

Reserve Remaining Network

192.168.1.112 – 192.168.1.255

for future departmental expansion.

This VLSM approach minimizes address wastage while supporting future scalability.

7. Optimized Configuration

Department	Optimized Subnet	Usable Hosts
Large Department	192.168.1.0/26	62
Medium Department	192.168.1.64/27	30
Small Department	192.168.1.96/28	14
Reserved Address Space	192.168.1.112 – 192.168.1.255	144 Addresses

8. Final Result

Subnet Allocation Category	Network Range / Details	Usable Hosts
Original Network	192.168.1.0/24	254 Hosts Total
Largest Subnet	192.168.1.0/26	62 Usable Hosts
Medium Subnet	192.168.1.64/27	30 Usable Hosts
Small Subnet	192.168.1.96/28	14 Usable Hosts
Remaining Unallocated Space	192.168.1.112 – 192.168.1.255	Reserved for Expansion
Optimization Technique	Variable Length Subnet Masking (VLSM)	Maximized Utilization

9. Analysis & Justification

The subnet allocation has been analyzed and verified successfully. No subnet overlap or addressing conflict exists in the current allocation; however, inefficient utilization may occur when subnet sizes do not match departmental requirements. Implementing Variable Length Subnet Masking (VLSM) allows subnet sizes to be allocated according to the actual number of devices in each department. This approach minimizes IP address wastage, improves address utilization, supports future network expansion, and simplifies network management while maintaining efficient routing.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

1. Problem Identification

Two network devices are configured with the IPv6 addresses 2001:db8::ff00:42:8329/64 and 2001:db8:0:0:1::1/64. Although both devices are physically connected to the same network, they cannot communicate. The objective is to analyze the IPv6 configuration, verify the network prefixes, identify the cause of communication failure, optimize the addressing configuration, and calculate the available host addresses within the subnet.

2. Given Configuration

Device	Configured IPv6 Address / Prefix
Device A	2001:db8::ff00:42:8329/64
Device B	2001:db8:0:0:1::1/64
Prefix Length	/64

3. Error Analysis

Expanded Address of Device A

2001:0db8:0000:0000:0000:ff00:0042:8329

Expanded Address of Device B

2001:0db8:0000:0000:0001:0000:0000:0001

Network Prefix of Device A

2001:0db8:0000:0000::/64

Network Prefix of Device B

2001:0db8:0000:0000::/64

Both devices belong to the same /64 network.

Therefore, there is no network prefix mismatch.

The communication issue is likely caused by incorrect IPv6 interface configuration, Neighbor Discovery failure, Duplicate Address Detection (DAD), or gateway configuration.

4. Debugging Process

Step 1

Expand both IPv6 addresses.

Completed successfully.

Step 2

Compare the first 64 bits.

Both prefixes are identical.

Step 3

Verify subnet membership.

Device A

2001:db8::/64

Device B

2001:db8::/64

Both belong to the same IPv6 subnet.

Step 4

Identify the possible fault.

No addressing mismatch exists.

Possible issues include:

- Incorrect IPv6 interface configuration.
- Neighbor Discovery Protocol (NDP) failure.
- Duplicate Address Detection (DAD) failure.
- Incorrect default gateway configuration.

5. Step-by-Step Calculation

IPv6 Address Length

128 bits

Prefix Length

64 bits

Host Bits

$128 - 64$

$= 64$

Available Host Addresses

2^{64}

$=$

18,446,744,073,709,551,616

6. Optimization

Assign simplified IPv6 addresses while maintaining the same /64 network prefix.

Recommended Device A

2001:db8::100/64

Recommended Device B

2001:db8::101/64

Network Prefix

2001:db8::/64

Verify IPv6 interface status and Neighbor Discovery operation.

7. Optimized Configuration

Parameter / Field	Optimized Value
Device A IPv6 Address	2001:db8::100/64
Device B IPv6 Address	2001:db8::101/64
Network Prefix	2001:db8::/64
Available Host Addresses	18,446,744,073,709,551,616

8. Final Result

Parameter	Final Output Value
Device A Prefix	2001:db8::/64
Device B Prefix	2001:db8::/64
Same Subnet	Yes
Network Prefix Mismatch	No
Possible Configuration Error	IPv6 Interface / NDP / DAD
Optimized Device A	2001:db8::100/64
Optimized Device B	2001:db8::101/64
Available Host Addresses	18,446,744,073,709,551,616

Possible Configuration Error IPv6 Interface / NDP / DAD

Available Host Addresses 18,446,744,073,709,551,616

9. Analysis & Justification

Both IPv6 devices are configured with the same /64 network prefix, confirming that they belong to the same subnet. Since there is no prefix mismatch, the communication failure is most likely related to IPv6 interface configuration, Neighbor Discovery Protocol (NDP), Duplicate Address Detection (DAD), or gateway configuration. Assigning valid IPv6 addresses within the same network and verifying the IPv6 interface settings ensures reliable communication. The corrected subnet provides 2^64 possible host addresses, making it suitable for modern enterprise and IoT environments.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

1. Problem Identification

A router maintains routing entries for multiple destination networks. Users report that packets destined for the IP address 192.168.2.100 are not reaching the intended network. The objective is to debug the routing configuration by identifying the matching routing entry, determining the next-hop router, verifying the destination subnet, locating any routing configuration errors, optimizing the routing table, and identifying the default route used when no matching destination exists.

2. Given Configuration

Destination Network	Next Hop Router
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1
Destination Packet IP	192.168.2.100

3. Error Analysis

Destination IP

192.168.2.100

Subnet Mask

255.255.255.0 (/24)

Applying the subnet mask

192.168.2.100

AND

255.255.255.0

Result

192.168.2.0

Therefore,

the destination IP belongs to the subnet

192.168.2.0/24

4. Debugging Process

Step 1

Verify the destination network.

Destination IP

192.168.2.100

belongs to

192.168.2.0/24

Step 2

Search the routing table.

Matching Route

192.168.2.0/24

Step 3

Apply Longest Prefix Match.

The router selects the route

192.168.2.0/24

because it is the most specific matching entry.

Step 4

Determine the next-hop router.

Next Hop

10.0.0.3

Step 5

Check for routing errors.

The routing table contains the correct route for the destination network.

No routing table mismatch is identified based on the given information.

If packets still fail, possible causes include:

- Incorrect next-hop configuration.
- Interface failure.
- Link connectivity issue.
- Access Control List (ACL) blocking traffic.
- Incorrect static route configuration on neighboring routers.

5. Step-by-Step Calculation

Destination IP

192.168.2.100

Subnet Mask

255.255.255.0

Network Address

192.168.2.0

Matching Route

192.168.2.0/24

Selected Next Hop

10.0.0.3

6. Optimization

Maintain the existing routing entry for

192.168.2.0/24

Verify that next-hop router

10.0.0.3

is reachable and operational.

Ensure all router interfaces are active and correctly configured.

Verify neighboring routers for consistent routing information.

Retain the default route for unknown destinations.

7. Optimized Configuration

Destination Network	Optimized Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route (0.0.0.0/0)	10.0.0.1

10.0.0.2

Destination Network

192.168.2.0/24

Next Hop

10.0.0.3

Default Route

0.0.0.0/0

Next Hop

10.0.0.1

8. Final Result

Parameter	Final Routing Output Value
Destination IP	192.168.2.100
Destination Subnet	192.168.2.0/24
Matching Route	192.168.2.0/24
Longest Prefix Match	192.168.2.0/24
Selected Next Hop	10.0.0.3
Routing Error	No Route Mismatch Detected
Optimized Default Route	0.0.0.0/0 via 10.0.0.1

Routing Error No Route Mismatch Detected

9. Analysis & Justification

The destination IP address 192.168.2.100 correctly belongs to the network 192.168.2.0/24. Using the Longest Prefix Match algorithm, the router selects the routing entry for 192.168.2.0/24 and forwards the packet to the next-hop router 10.0.0.3. Since the routing table already contains the correct entry, the communication problem is likely caused by external factors such as interface failure,

link connectivity, ACL restrictions, or neighboring router configuration rather than an incorrect routing table entry. The default route (0.0.0.0/0 via 10.0.0.1) will be used only when no specific destination network matches.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation